Procurement: the track unit on Nhien’s Rev B frame

Date 16 Oct 2026, updated 19 Oct (3 x 5 rail screws, stage-2 pins 2.5 x 12). Prices S$ ex GST as seen in listings; confirm at checkout. Rev B = Nhien’s pack (`RB4107 Rev B`), whose BOM covers everything in section 2. The track comes in two options, A and B; everything else is the same whichever is chosen, so section 2 and the pulleys can be ordered first.

1A. Track, option A: my printed track

Part	Qty	Make / buy	Where	Price	Arrives
Track, TPU 95A, `stl/track_TPU_x4.stl` (drawing TS-01)	4	**Make**: Fab Lab UP300 (TPU head)	Element14 SG TPU 95A filament if the lab has none	~S$45/kg, uses 130 g	filament 1-2 days; 3 h print each
Fit-test slice, `stl/fit_test_track_5mm_TPU.stl`	1	Make	same	-	10 min
Stand-in pulley, `stl/stand_in_pulley_20T_PLA_test_only.stl`	1	Make, PLA	any printer	-	1 h

Deck: no change needed; the 3 mm cleats sit 5 mm under the deck.

1B. Track, option B: Nhien’s bought belt with cleats (Rev B B10)

Part	Qty	Make / buy	Where	Price	Arrives
Track belt HTD 550-5M-15, 110 teeth, 15 wide (e.g. SKF PHG 550-5M-15), drawing TS-01B	4	**Buy**	RS SG, stock no. 058-6535 or MISUMI SG HTD 5M belts	see listing (about S$25-35 each)	RS SG stock: 1-3 days
Cleats, rubber 10 x 7.9 x 15, bonded on the belt back every 5th tooth (25 mm), 22 per belt	88	**Buy + bond**: belt shop	Local belt / conveyor shop (Fenner, Megadyne or Gates distributor), with the belts	quote	allow 1-2 weeks
Deck change for the belt: notch P12 13 x 135 over each front lane (x -25 to 110), or cut the cleats to 5 high	1	**Make**: with the deck laser cut, or file / mill after	Professional Workshop	-	with the deck

1C. Wheels, both options

Part	Qty	Make / buy	Where	Price	Arrives
Timing pulley HTD 5M 20T, 15 mm belt, flanged (19 Oct: MISUMI economy HTD 5M, code **H5M**, not S5M). **Hinge wheels: 4 + 1 with a finished Ø8 bore**: shaft-hole type H (plain round, no tapped hole) if the configurator offers it for 20T, else P (round with tapped hole, ignore the screw; pin hole 90° from it). **Idlers: 4 with the smallest bore offered**, bored to Ø16 at the workshop (20T stops at about Ø15, so no 16 bore). Before ordering, check on the drawing: width over the flanges 19 or less (the lane is 20) and no hub (shape B has a Ø19 hub)	8 + 1	**Buy**	MISUMI SG, Timing Pulleys HTD 5M	~S$20 each	ships 1 day from SG stock; allow 2-5 days
Pulley bore modification: 4 x bore Ø16 H7 (idlers, M2) only; the hinge wheels come bored Ø8. Stage-2 pulleys M3 (Nhien's): ream Ø8 H7. Ø2.5 cross holes drilled on our drill press in a V-block, shaft drilled through them at assembly (TS-06, TS-11)	4 + 2	**Make**: Professional Workshop request	E3-01-06	free	workshop queue, allow 1-2 weeks
Spring pins ISO 8752 2.5 x 24 (hinge pulleys, replaces E-clips F22 and the D-flats)	4 + 4 spare	**Buy**	RS SG spring pins	~S$5 / pack	RS SG stock
Spring pins ISO 8752 2.5 x 12 (stage-2 pulleys M3 only, if Nhien pins them (Ø17.5 OD, tooth roots about Ø13.4: a longer pin would reach into the belt grooves))	2 + 2 spare	**Buy**	RS SG spring pins	~S$5 / pack	RS SG stock

Part	Qty	Make / buy	Where	Price	Arrives
Set screws M4 x 5 cone point DIN 914 (stage-2 pulleys M3 only, if Nhien keeps his pulley's tapped hole: cone point into a seat spot-drilled in the shaft through that hole, Loctite 243; the hinge wheels have no hub in the 20 mm lane, so they stay pinned; plan Appendix C item 17)	2 + 2 spare	**Buy**	RS SG set screws	~S$3 / pack	RS SG stock
Hinge shaft: plain round Ø8 h7 ground steel, 300 mm (100 mm to practise on) (replaces the D-shaft B13; cut to 82.3 and 77.3, no grooves)	1	**Buy**, cut	RS SG linear shafts or MISUMI SG PSFJ8	~S$10	RS SG stock

1D. Tension and rail support (from the design review), both options

Part	Qty	Make / buy	Where	Price	Arrives
Tensioner tab, Al angle 20 x 20 x 3 cut 30 with a 7.5 x 8 ear, M4 tapped (TS-09)	4	**Make**: hacksaw, drill, M4 tap	from BOM B16 angle (order 300 mm instead of 100)	-	1 h
Tensioner slider, an 8 mm slice of the same angle, leg 19 with Ø6.4, foot 8 (TS-09; replaces the M6 washer under the nut)	4	**Make**: hacksaw, drill	from the same angle (order 400 mm)	-	30 min
Thread-forming screws 3 x 5 pan head for plastics (rail to carrier angle, 3 per rail; 3 x 5 not 3 x 10: the rail is 5 thick)	24 + 6 spare	**Buy**	RS SG thread-forming screws	~S$6 / pack	RS SG stock
M3 x 6 SHCS ISO 4762 (carrier angles to plates 24, tensioner tab ears 4)	28 + 4 spare	**Buy**	RS SG	~S$5 / pack	RS SG stock
Jack screw M4 x 20 socket head + M3 x 6 for the tab	4 + 4	**Buy**	RS / local	-	stock
New M3 tapped holes in P1 and P2: (x +-223, y +11) for the tab (TS-09) and six at (x +-40, +-105, +-170; y +-3.5) for the carriers (TS-10)	4	**Make**: add to the DXF, or drill 2.5 + tap M3 on site	with the plates	-	with the plates
Rail carrier angle, Al 10 x 10 x 1.5 x 186, plate leg cut to 6.5, 3 + 3 holes (TS-10; replaces the 32 RS 427-991 clips)	8	**Make**: from 1.6 m of angle	local metal supplier / MISUMI SG	~S$10	1-3 days

Part	Qty	Make / buy	Where	Price	Arrives
Rails P3-P6 trimmed from 189 to 186 at the idler end (TS-10, TS-11)	8	Make: hacksaw	-	-	10 min

2. Nhien's (frame, drive, hinge, electronics), from the Rev B BOM

Part	Qty	Make / buy	Where
Side plates P1 (front) and P2 (rear), 5 mm Al, laser + tap	2 + 2	Make	DXF in the pack; Professional Workshop or a laser shop on the vendor list
Slide rails P3-P6, UHMW strip 189 x 20 x 5	8	Buy + drill	MISUMI SG UPA-189-20-5
~~Rail brackets RS PRO 427-991~~: **do not order**, replaced by the TS-10 carrier angles (1D)	0	-	-
Hinge shafts P20 A 82.3 + B 77.3, now plain round Ø8 h7 (section 1C), no D-flat, no E-clip grooves	2	Buy stock, cut	see 1C
Sector gear P11: proposed KHK SSAY1-100 (module 1, 100 teeth, 20°, face 6, bore 10, S45C, flat plate) cut to a 23-tooth sector (plan 5.14); needs Nhien's OK on the centre distance 60.1 to 60.2 (P13 slot 1 mm closer) or a 22-tooth module 1 pinion in place of M6	1	Buy, cut	KHK catalog (thin-face hubless SSAY series, about US$43) or MISUMI SG (KHK brand), search SSAY1-100; plan B any 1M 100T 45# steel gear from Shopee / AliExpress, faced to 6 mm at the Professional Workshop
Bearing 688-2RS 8 x 16 x 5	14	Buy	RS SG ball bearings
Shoulder screw ISO 7379 Ø8 x 25 M6 + washers, shim, spacer, nyloc	4 sets	Buy	RS SG shoulder screws
2020 profile x 110 (P7) x2, 3030 x 160 (P8) x2, corner posts P15 x4	1 m + 0.5 m	Buy, cut	MISUMI SG / local
Corner brackets: standard cast 2020 and 3030 inside corner brackets instead of the P18 / P19 angle offcuts (bigger contact area, design review); T-nuts, shims 8x11x0.5 and 8x11x1; E-clips F22 no longer needed	-	Buy	MISUMI SG / local
Deck P12 (slot H 3 mm longer, see pack page 2), tray P16, cover P22, motors, drivers, ESP32, battery	-	Theirs	Rev B BOM
Bearing seats in P1, P2, P17: laser-cut Ø15.8, reamed Ø16 H7 on site, 688-2RS retained with Loctite 641 (TS-11)	6	Make	Professional Workshop, 16 H7 reamer

3. Choosing between A and B

Option A is in-house: four 3 h prints, no supplier, 5 mm deck clearance, and the cleat spacing is matched to the ES1 step. Option B is bought: belts from RS in days, but the 88 cleats need a belt shop (1-2 weeks) and the deck needs a notch or shorter cleats. Wheels, bore modification, plates, shafts, bearings and shoulder screws are the same in both.

4. Order of events (19 Oct)

1. Mon 19 to Tue 20 Oct: check the MISUMI H5M 20T 150 drawing (19 or less over the flanges, no hub), then order the pulleys (4 + 1 bored 8, 4 idlers smallest bore), the RS order (shaft, pins, screws, jack screws) and the aluminium angle. Print the fit-test slice and stand-in pulley (option A).
2. 22 to 29 Oct: Nhien re-issues the P1 / P2 DXF and has the plates laser cut. As parts arrive, make the tabs, sliders and carrier angles, trim and drill the rails, drill the hinge wheels' Ø2.5 holes.
3. When the idlers arrive (about 23 to 26 Oct): one workshop request for the idler bores (16 H7), the stage-2 pulley reams (8 H7) and, once the plates are back, the bearing seats (16 H7). Allow 1 to 2 weeks.
4. 9 to 10 Nov: assemble; 11 to 13 Nov: tension and run tests; 16 Nov: advisor demo.